

LLM Copy-Paste Residue Check

Scan report for the active `thesis.tex` dependency tree

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Scope

This report checks the active dissertation source for phrases that often appear when text has been pasted from an LLM chat or from generic LLM-style rewrites. The check covers the active `thesis.tex` dependency tree only. Existing thesis files were not modified.

This is not an AI-authorship detector. The goal is narrower and more useful: find chat residue, prompt/answer residue, generic filler, tutorial tone, and source scaffolding that should be removed before final submission.

Summary

- **No direct smoking-gun chat leakage found.** I did not find active occurrences of phrases such as `here is`, `as requested`, `according to your definition`, `based on your definition`, `using your definition`, `your thesis`, or `let me`.
- **Some false-positive direct matches were harmless.** For example, `chapterconflict.tex:529` and `forwardtight.tex:21` use the word “below” in normal academic prose.
- **Real issues exist in three clusters:** prompt/response residue in the introduction’s LLM example, generic/tutorial phrasing in several technical sections, and source-scaffolding language around the framework-user discussion.
- **Highest-priority cleanup:** review `einleitung.tex:67–81`, `chapterconflict.tex:107`, `chapterconflict.tex:1502`, `tightmotivation.tex:673–738`, `qviol.tex:868`, and `fazit.tex:10–38`.

Pattern Counts

After stripping fully commented lines, the scan found the following counts:

Category	Matches
Direct chat leakage	2
Prompt/response residue	13
Generic transitions	24
Vague filler	17
Hedging	5
Instructional/tutorial tone	18
Scaffolding or source debris	2
Citation/fact red flags	23

The direct-chat count is misleading: both matches were ordinary uses of “below”, not actual chat leakage.

Priority 1: Inspect and Rewrite

1. **Prompt/response wording in the introduction.** `einleitung.tex:67` and `einleitung.tex:81` discuss a prompt, a response, and a hallucination detector. This may be intentional because the passage is about LLM hallucinations, but it reads close to chat-analysis language. Suggested action: make it explicit that this is a documented case study or demonstration, not an embedded chat transcript. Avoid vague phrases such as “The solution uses” unless the solution is named.
2. **“The user” phrasing in formal framework text.** `chapterconflict.tex:107`, `chapterconflict.tex:14`, `chapterconflict.tex:1502`, and `conflictmeat.tex:1180` refer to “the user” of the framework. This is not necessarily wrong, but it sounds like tool-documentation prose rather than dissertation prose. Suggested action: replace with “the designer”, “the analyst”, “the regulator”, or “the specification engineer”, depending on context.
3. **Broken framework-user paragraph.** `chapterconflict.tex:1502` combines user/tool language, generic terms, and an already-known broken sentence: “Unless all the other tools in the framework are.” This is one of the strongest copy/paste-scaffolding signals in the scan. Suggested action: rewrite the paragraph from scratch as an academic workflow.
4. **Tutorial examples with “The user inputs”.** `tightmotivation.tex:673-738` repeatedly uses phrases such as “The user inputs”. This is acceptable for interactive examples, but it reads more like a generated tutorial than a formal dissertation section. Suggested action: use “the trace begins with”, “the next symbol is”, or “the input word is extended by”.
5. **Question-answer setup in quantitative blame.** `qviol.tex:868` asks: “does adding up the two blame counters always give back the same score”. This conversational question style is not a direct LLM leak, but it reads like explanatory chat prose. Suggested action: rewrite as a formal motivation for the lemma and align it with the actual inequality proved.
6. **Conclusion still has generic LLM-like phrasing.** `fazit.tex:10`, `fazit.tex:20`, and `fazit.tex:38` contain phrases such as “clear and practical”, “more expressive and easier to use”, and “comprehensive solution”. These are plausible but generic. Suggested action: replace with concrete claims about decidability, finite-state monitor synthesis, conflict density, and blame attribution.

Priority 2: Citation and Fact-Claim Red Flags

These are not LLM leaks, but they are common LLM-style overclaims unless they are properly sourced or carefully qualified.

Location	Flagged wording	Suggested action
<code>einleitung.tex:29</code>	“Studies show that in places with many regulations...”	Add a citation or make the claim more cautious.
<code>einleitung.tex:33</code>	“in the literature”, “widely cited”, “roughly”	The footnotes help; keep exact source framing and avoid sounding broader than the cited reports.
<code>einleitung.tex:43</code>	“According to Legal Complex... hundreds of millions”	The figure source helps; consider adding access date and exact scope.
<code>chapterconflict.tex:404</code>	“well known procrastination effect”	Already cited; keep if the cited works explicitly support this term.

<code>chapterconflict.tex:660</code>	“commonly used”	Harmless, but can be shortened.
<code>chapterconflict.tex:688-690</code>	“According to our definition(s)”	Not chat leakage, but repetitive; replace with direct formal reference.
<code>conflictmeat.tex:252</code>	“According to our definition”	Fine technically; consider “By Definition ...”.
<code>qviol.tex:114,145</code>	“According to the definition...”	Fine technically; “By Definition ...” is cleaner.

Priority 3: Generic or Tutorial Tone

These passages are probably harmless, but they are places where the prose may sound generated or overly didactic.

- `chapterconflict.tex:696`: “We saw that... We proceed now to show...” could be tightened to “We now prove that...”.
- `chapterconflict.tex:774`: “We now turn to...” is normal but repeated often. Use sparingly.
- `conflictmeat.tex:958`: “We now show... very natural normative intuitions” sounds conversational. Prefer a direct theorem-introduction sentence.
- `chaptermonitor.tex:11`: “We begin by motivating...” is normal, but the paragraph already has several guidepost phrases. Reduce navigation prose.
- `chaptermonitor.tex:348`, `chaptermonitor.tex:1196`, `tightmotivation.tex:425`, `tightmotivation.tex:751`: repeated “Crucially” can sound LLM-like. Keep only where the claim is genuinely a turning point.
- `forwardtight2.tex:200`: “Let us consider instead...” is fine in a proof/example, but use consistently with the rest of the thesis style.
- `quantitativeviol.tex:463`: “We now turn to...” is acceptable but can be shortened.

Probably Harmless Matches

- `chapterconflict.tex:529`: “The satisfaction relation below...” is ordinary exposition.
- `forwardtight.tex:21`: “semantic clauses below” is ordinary exposition.
- `preliminaries.tex:51`, `preliminaries.tex:276`, and `preliminaries.tex:424`: “Consider the following examples” is normal in mathematical writing.
- `declaration.tex:13-14`: AI tools are explicitly declared. This is not leakage; it is required disclosure text.
- `einleitung.tex:67-81`: “prompt” and “response” may be necessary because the passage discusses an LLM hallucination example. It still deserves stylistic cleanup.

Recommended Manual Search Patterns

Before final submission, run quick searches for these exact phrases:

here is
here are
as requested
according to your definition
based on your definition
using your definition
your thesis
your framework
the user
the prompt
the response
step-by-step
it is important to note
this highlights
this underscores
serves as a
provides a robust
clear and practical
comprehensive solution
roughly speaking
we now show
we now turn to
let us consider
TODO
placeholder
citation needed

Bottom Line

The scan did not find obvious active chat-copy residue such as “here is the rewritten paragraph” or “according to your definition”. The dissertation does, however, contain several passages that read like generated explanatory prose or tool documentation. The highest-value cleanup is to replace “the user” with a domain role, rewrite the broken framework workflow paragraph, reduce tutorial phrasing in tight-semantics examples, and make broad claims more precise or cited.